Canadian Natural Gas Production Trends
and
Upstream Capacity to Support LNG Canada Phase 1

Data Science Mentorship Program – Society of Petroleum Engineers
Calgary Section

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1. Executive Summary

Canada's natural gas sector has experienced significant structural changes over the past two decades, driven by shale development, infrastructure constraints, and the emergence of LNG[i] export projects.

This report evaluates whether historical production growth and recent market dynamics are sufficient to support future LNG export capacity, with a focus on whether incremental supply can generate the surplus required for LNG Canada Phase 1. Forecasts indicate that production is expected to remain relatively stable through 2030.

The present analysis is limited to upstream production projections and does not incorporate midstream transportation capacity or downstream liquefaction constraints. Therefore, the findings reflect production capability rather than fully integrated export feasibility.

2. Data & Methodology

The analysis uses monthly marketable natural gas production data from the Canada Energy Regulator (CER), covering marketable natural gas production from 2000 to 2025. The data reported in both million cubic feet per day (mmcfd) and thousand cubic meters per day (10^3 m^3/d).

For this study, **thousand cubic meters per day (10^3 m^3/d) is adopted as the primary analytical unit**, as it aligns with Canadian regulatory reporting practices and facilitates comparison with international natural gas markets. Monthly observations are aggregated into annual averages and further converted into a consistent unit of **million tonnes per annum (mtpa)** to support long-term trend analysis and LNG-related comparisons.

The study is structured into five phases: data preparation and unit harmonization, historical production trend analysis, structural change and growth regime analysis using rolling windows, LNG export readiness assessment, and short-term production forecasting.

3. Phase 0: Data Preparation & Unit Harmonization

To ensure the validity and comparability of the analysis, the raw production data were systematically cleaned, restructured, and standardized. The original dataset contains inconsistencies in units, temporal frequency, and formatting across years. These issues were addressed through unit harmonization, time alignment, and aggregation, resulting in a consistent and analysis-ready dataset used in all subsequent phases.

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4. Phase 1: Annual Production & YoY Growth

In this phase, we examine the long-term evolution of Canadian marketable natural gas production from 2000 to recent years. The analysis focuses on overall production trends, growth dynamic including measured by Year-over-Year (YoY) changes and the Compound Annual Growth Rate (CAGR), and the identification of periods of growth, stagnation, and decline.

The historical patterns are further interpreted in the contextual of major energy market developments, including oil/gas price cycle, the U.S. shale revolution[1][2], the COVID-19 pandemic[3], and the key milestones surrounding LNG Canada's final investment decision (FID)[4].

4-1. Long-term Trend (2000 → Present)

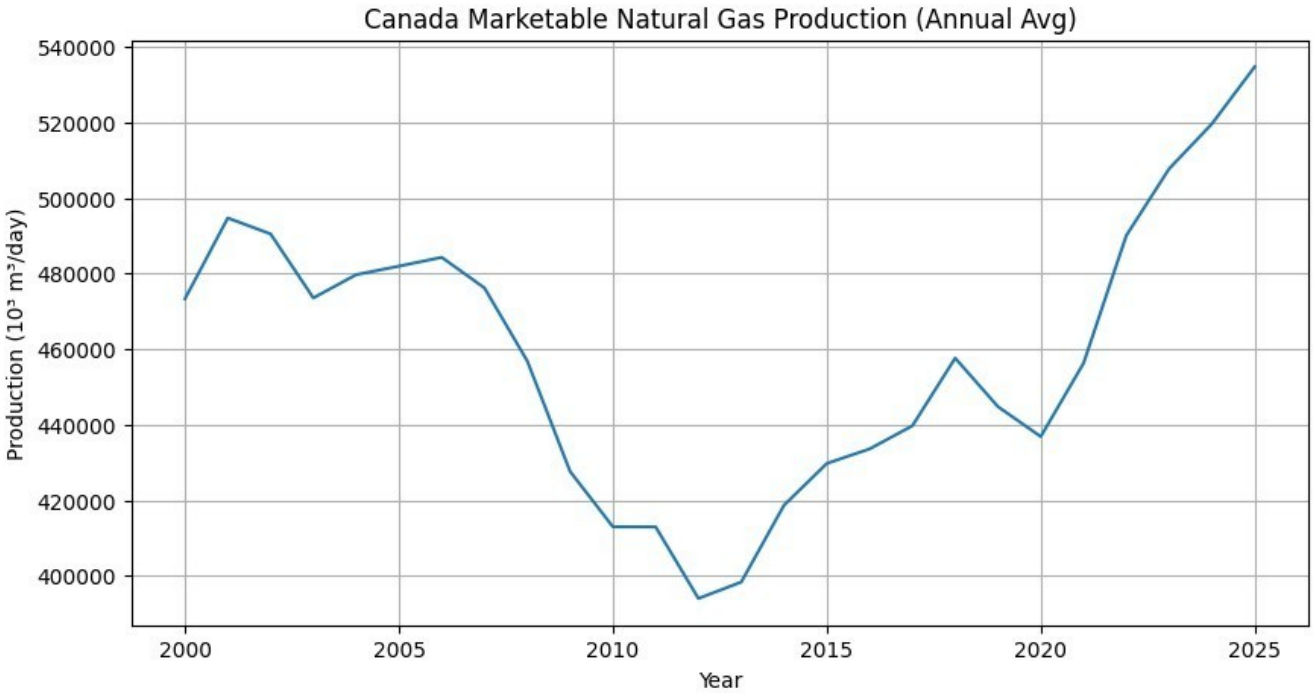

Figure 1: Canada Marketable Natural Gas Production (Annual Avg)

We first examine the long-term production trend of marketable natural gas in the Canadian market to understand the overall growth momentum and fluctuations.

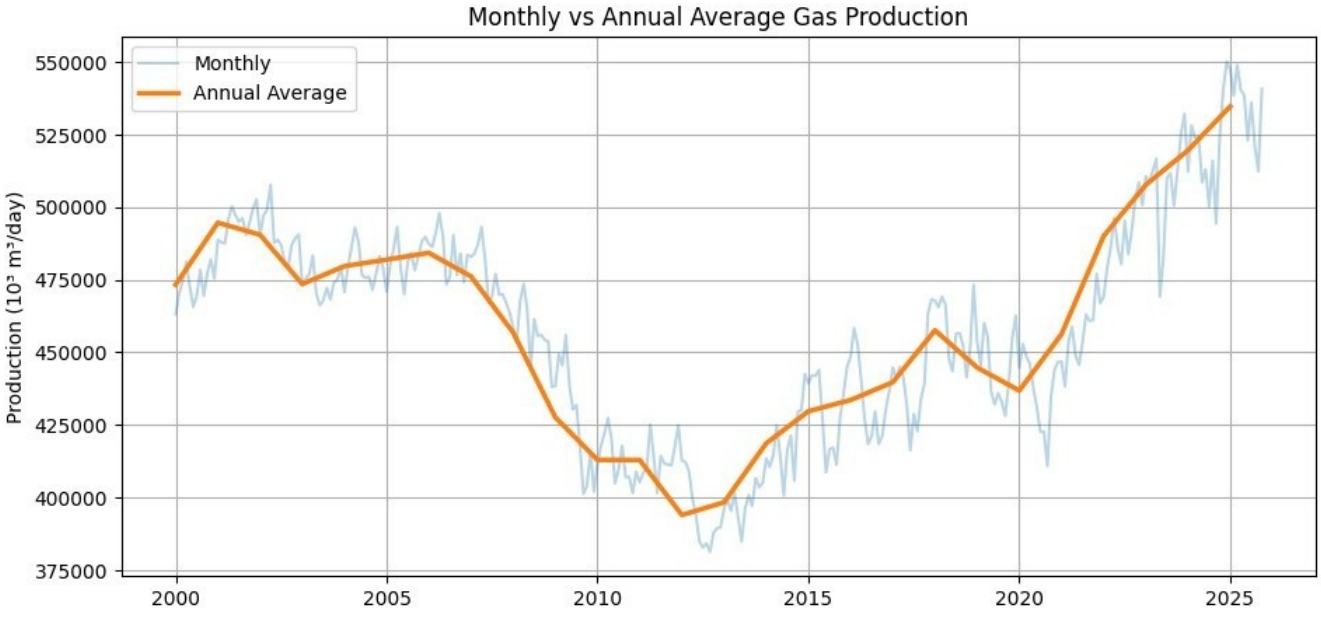

Figure 2: Monthly vs Annual Average Gas Production

We also retain monthly fluctuations and yearly trends to better examine the overall volatility in production.

4-2. Growth Dynamics

4-2-1. Year-over-Year (YoY) Growth

Year-over-Year (YoY) growth measures the percentage change of a variable compared with the same period in the previous year. It helps identify whether production increased or decreased each year while removing seasonal effects.

For this analysis, we used **a ±3% threshold** to classify each year into three regimes:

- Growth (YoY > 3%)
- Decline (YoY < -3%)
- Stagnation/Plateau (-3% ≤ YoY ≤ 3%)

The resulting plot is shown below, highlighting the annual ups and downs in Canadian marketable gas production.

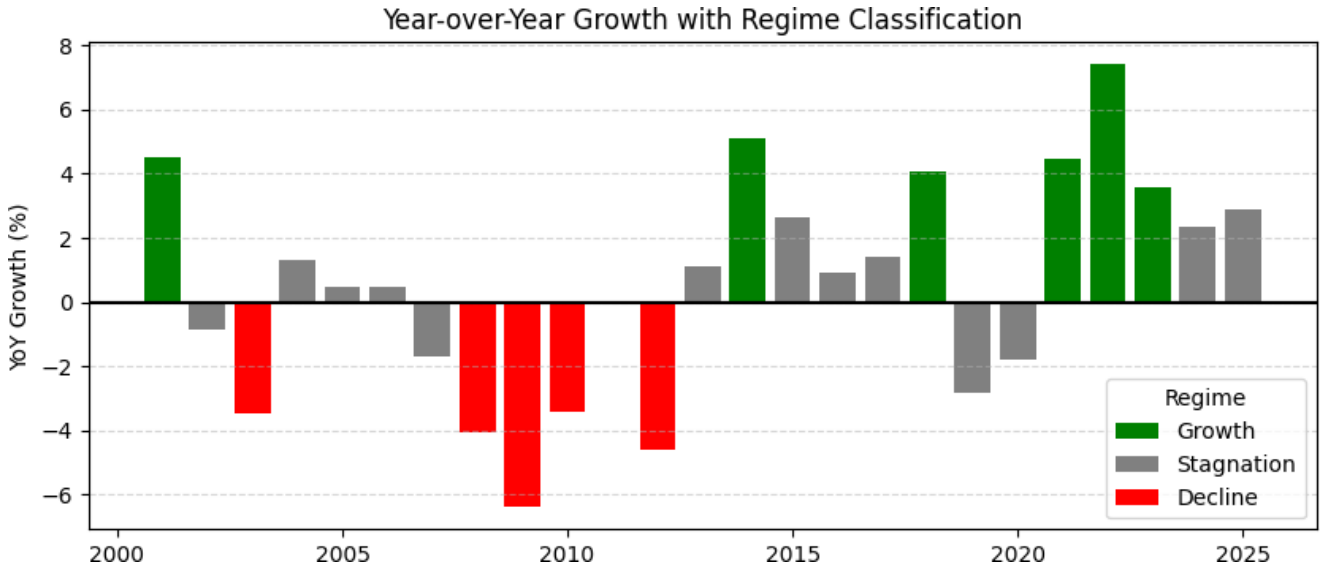

Figure 3: Year-over-Year Growth with Regime Classification

From the mid-2000s, Canadian marketable natural gas production experienced a decline, starting around 2006–2007 and continuing through roughly 2012. During this period, the YoY growth plot shows multiple years of negative growth, reflecting both the oil and gas price cycles and the lingering effects of the 2008 financial crisis. Production began to recover after 2012 as the North American shale gas revolution matured, increasing supply.

Around 2020, production was relatively flat, likely impacted by COVID-19 disruptions and temporary market uncertainty. Since then, production has resumed a growth trend, coinciding with the 2018 final investment decision (FID) for LNG Canada and post-pandemic demand recovery.

Overall, the combination of long-term trend and YoY analysis highlights how structural shifts, market cycles, and major events shaped Canadian gas production over the past two decades.

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4-2-2. Compound Annual Growth Rate (CAGR)

The Compound Annual Growth Rate (CAGR) represents the average annual growth of a variable over a period of time, assuming growth compounds smoothly each year.

- CAGR provides **a single number** summarizing the overall long-term trend.
- It is especially useful to communicate the general direction of growth without being distracted by short-term fluctuations.

From 2000 to 2025, Canadian marketable natural gas production **grew slowly but steadily**, at an average rate of **0.49%** per year. While this may seem small on an annual basis, the cumulative growth over 25 years is significant.

Looking at individual years, 12 years experienced strong growth, 8 years saw declines, and 5 years remained relatively flat, as shown in the trend analysis plot. This illustrates that, despite short-term ups

and downs, **the long-term trend is a gradual and consistent increase**, influenced by market cycles and global events.

5. Phase 2: Rolling Trends & Structural Regimes

To reduce the impact of short-term year-to-year volatility and highlight long-term structural patterns, we apply a **3-year rolling average** to annual production data.

With this approach, each data point represents the average of the previous year, the current year, and the following year. This smoothing process helps filter out temporary shocks and better reveal underlying trends in production behavior.

5-1. Comparison: Preliminary EDA[ii] vs Smoothed Structural Analysis

5-1-1. Preliminary EDA (Yearly YoY Growth)

- This approach directly examines the **year-over-year percentage change** in production without any smoothing.
- **Results**:
    - 12 years showed strong growth
    - 8 years experienced decline
    - 5 years remained relatively stable
- **Interpretation**:

This method is highly **sensitive** to short-term movements, capturing sharp increases or decreases driven by one-off events, price shocks, or temporary market conditions.

It is useful for understanding **individual annual performance**, but can over-emphasize short-lived fluctuations.

5-1-2. Rolling Average & Structural Regime Analysis

- In this step, we first smooth the production series using a **3-year rolling average**, then classify each year into growth, decline, or plateau regimes.
- **Results**:
    - High-growth years decreased from 12 to 6
    - Declining years decreased from 8 to 5
    - Plateau / stable years increased to 15

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- **Interpretation**:

The rolling average **smooths out short-term noise**, meaning that isolated one-year spikes or drops no longer dominate the classification.

- Small, temporary increases or decreases are absorbed into the surrounding trend
- Only **persistent and structural shifts** are identified as growth or decline

This analysis highlights periods where production truly **changed direction over multiple years**, rather than reacting to short-term market volatility.

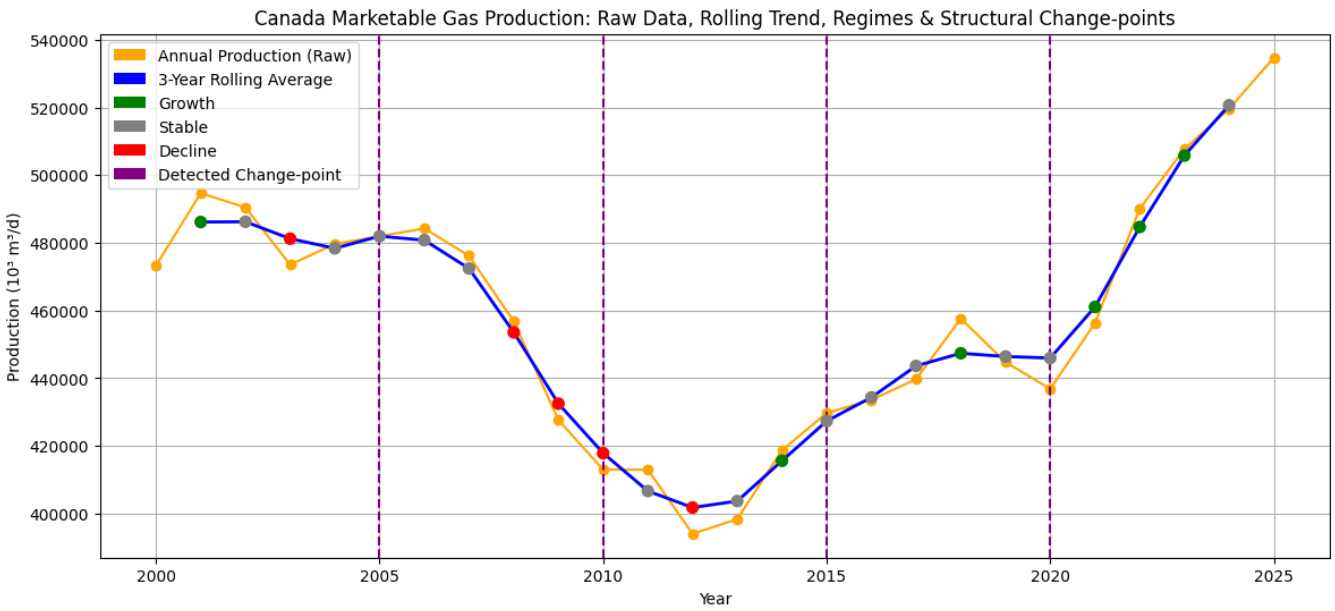

Figure 4: Canada Marketable Gas Production: Raw Data, Rolling Trend, Regimes & Structural Change-points

The detected change points line up with major shifts in the North American gas market.

Around 2005, production dynamics began to change as shale gas development started to influence supply growth.

By 2010, the U.S. shale gas boom had matured, increasing supply and putting pressure on prices, which altered Canada's production growth pattern.

Around 2015, production shifted from growth to a more stable plateau following the 2014–2016 oil and gas price collapse[5].

In 2020, another structural change occurred, reflecting the impact of COVID-19 disruptions and subsequent market restructuring.

6. Phase 3: LNG Canada Phase 1 Demand vs Supply Adequacy

Daily production data are useful for operational monitoring. However, LNG project demand and export capacity are typically discussed in **annual volumes**. To make Canada's gas supply directly comparable with LNG project requirements and global gas markets, production is converted **from thousand cubic meters per day (10^3 m^3/d) to million tonnes per annum (mtpa[6][7][8])** using an industry-standard LNG conversion factor, consistent with official reports.

6-1. Total Marketable Natural Gas Production Overview

LNG Canada Phase 1 is expected to require approximately **14 million tonnes per annum** of natural gas supply once fully operational. This figure represents the **annual feedgas requirement**.

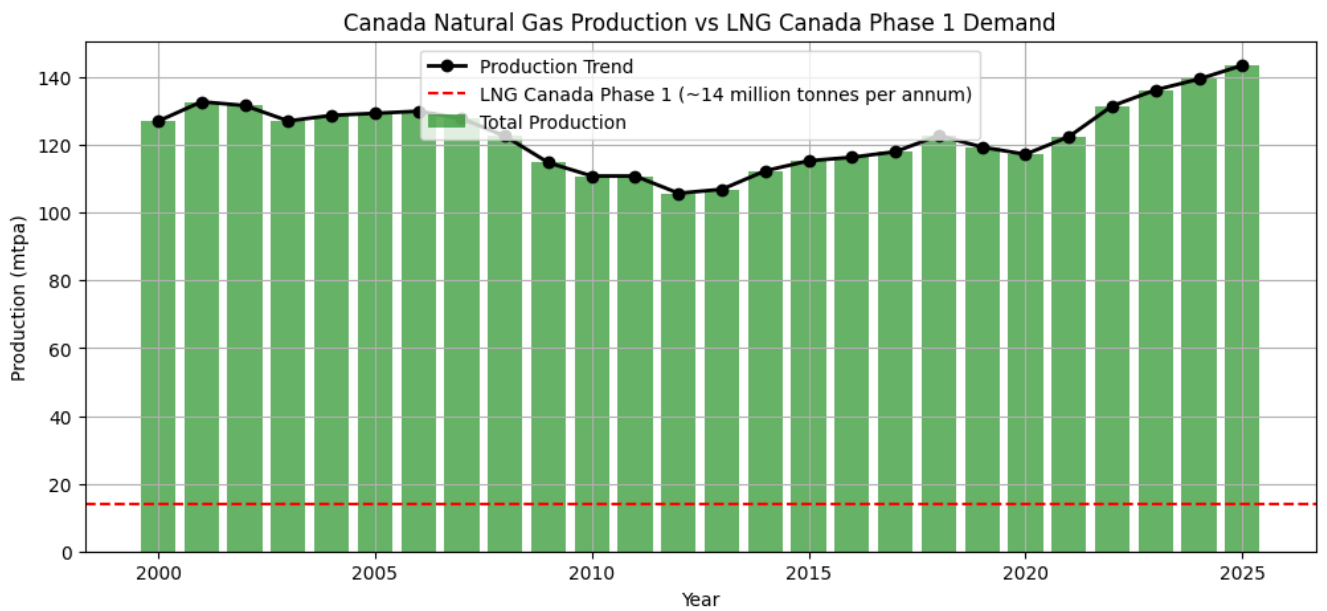

Figure 5: Canada Natural Gas Production vs LNG Canada Phase 1 Demand

At first glance, the volume of LNG demand appears small relative to Canada's total marketable gas production. However, total production does **not** necessarily represent available capacity for new LNG projects. In practice, existing output is largely absorbed by **domestic consumption** and **established pipeline exports to the United States**, indicating that the system operates near balance.

From a macro supply-demand perspective:

> *Production = Domestic Consumption + Exports + Storage Adjustment*

Therefore, the effective availability for LNG depends not on total production levels, but on **incremental surplus** within the system.

Under a balanced market structure, incremental availability can be approximated as:

> *ΔAvailable ≈ ΔProduction − Δ(Domestic + Exports)*

Given that domestic consumption is politically and economically sensitive, and pipeline exports are often supported by long-term contractual commitment, incremental LNG feedgas supply is primarily dependent on **sustained production growth**.

6-2. Incremental Supply Growth

While total production indicates overall capacity, LNG projects must ultimately be supported by incremental supply growth. To assess this, we examine year-over-year (YoY) changes in production, measured in mtpa.

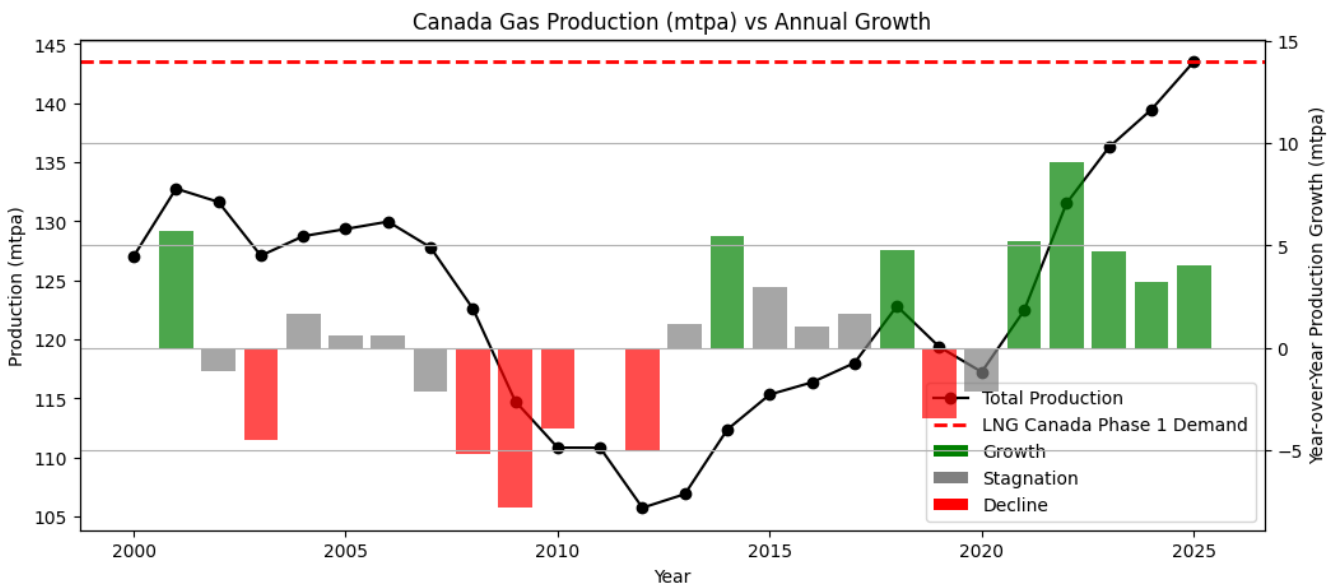

Figure 6: Canada Gas Production (mtpa) vs Annual Growth

The results indicate that annual production growth varies significantly across years and generally remains **below** the LNG Phase 1 demand threshold, suggesting that LNG supply would likely rely on **multi-year cumulative growth** and long-term production planning.

7. Phase 4: Forecasting Canada Marketable Gas Production

Following Phase 3 Analysis, several major structural breaks identified around:

- **2005** – Shale gas expansion in North America
- **2010** – North America production transition, coinciding with the reversal of the long-term decline in U.S. oil production (post-2009 shale expansion)
- **2015** – Post-oil price collapse adjustment
- **2020** – COVID-19 demand shock

These structural shifts suggest that Canadian marketable gas production experienced multiple regime changes over the past two decades.

To ensure a rigorous forecasting process, a consistent modeling framework is applied across all models, Prophet[11], and SARIMA[12]:

- **Training period**: 2000–2022
- **Validation period**: 2023–2025
- **Refitting period**: 2000–2025
- **Forecast horizon**: 2026–2030

Models are first trained on the historical sample (2000–2022) and evaluated using out-of-sample (2023–2025) to assess predictive performance and prevent overfitting. After validation, each model is re-estimated using the full dataset through 2025 to ensure optimal parameter estimation before generating the final forecast for 2026–2030.

The objective is to capture **seasonal variation** (Winter peaks vs Summer troughs) and assess whether these fluctuations could constrain LNG baseload operations. Outputs include the **monthly production path to 2030** and a **seasonal spread analysis**, with uncertainty framed via forecast ranges (95% CI[iii]). All production values are reported in **mtpa**.

7-1. Prophet

To improve sensitivity and control trend flexibility, we explicitly specify changepoint values and adjust the changepoint prior scale to 0.5. For clearer presentation, the forecast plots are cropped to focus on the relevant time window.

7-1-1. Validation

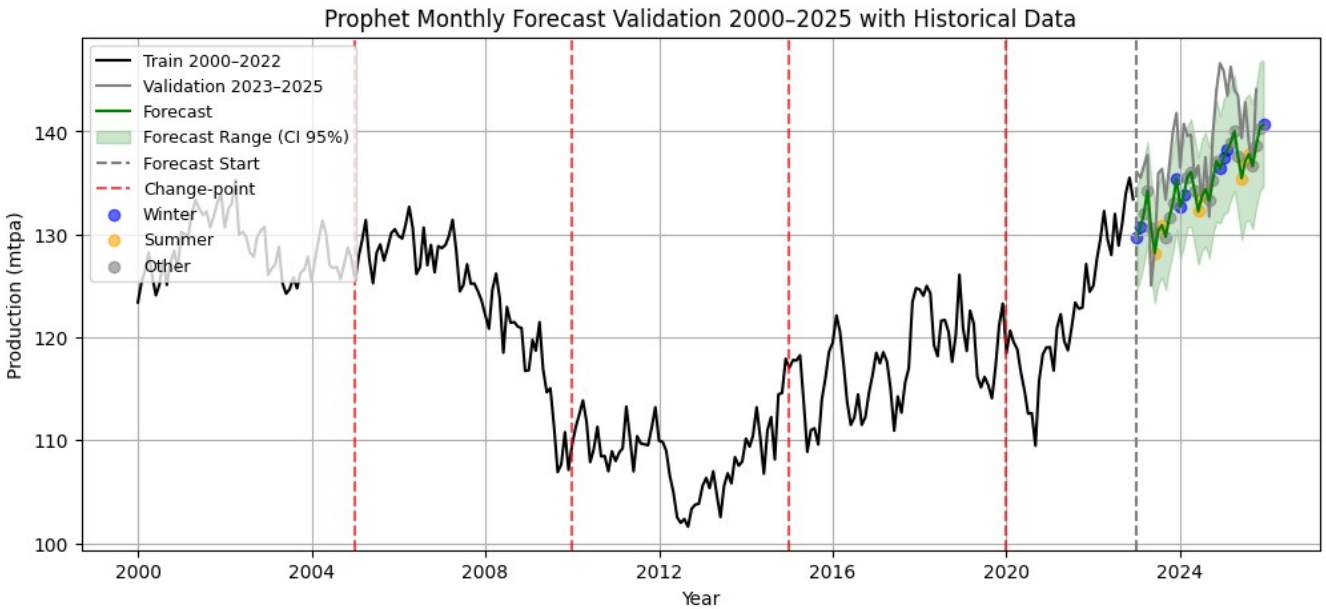

Figure 7: Prophet Monthly Forecast Validation 2000–2025 with Historical Data

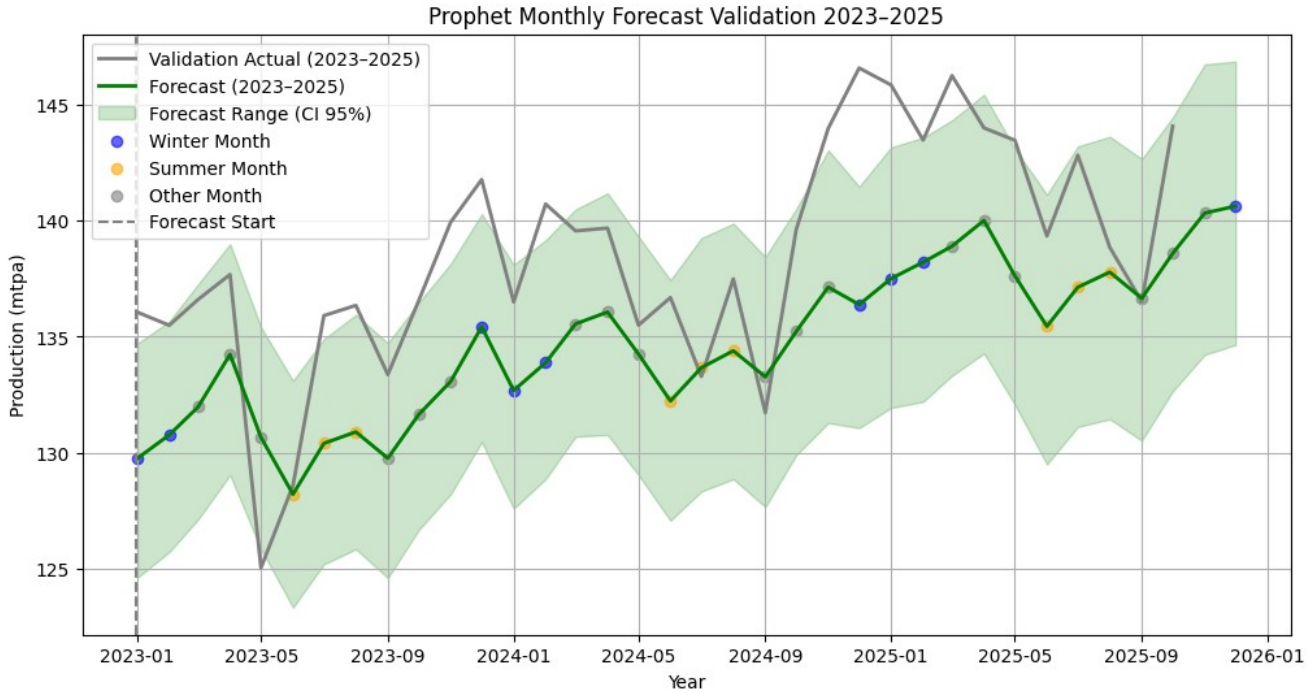

Figure 8: Prophet Monthly Forecast Validation 2023–2025

Figures 7 and 8 illustrate the Prophet model's learning and forecasting process for Canadian marketable gas production. Figure 7 presents the full period from 2000 to 2025, showing how the model captures long-term trends, structural regime shifts, and seasonal patterns. The historical data are shown in black, validation actual in gray, and forecast values in green, with the 95% confidence interval highlighted. Although the gray line (actual) is generally higher than the green line (forecast) in the short term, indicating a slight underestimation of recent shifts, the model effectively captures the trend direction and seasonality patterns.

Figure 8 zooms in on the validation period 2023–2025, providing a clearer view of short-term forecasting performance. It highlights how the model responds to seasonal peaks and troughs, enabling evaluation of forecast accuracy and uncertainty specifically for this period. The combination of these figures demonstrates that Prophet is capable of reflecting both structural changes and seasonal variability while providing a quantified range of uncertainty.

7-1-2. Forecast

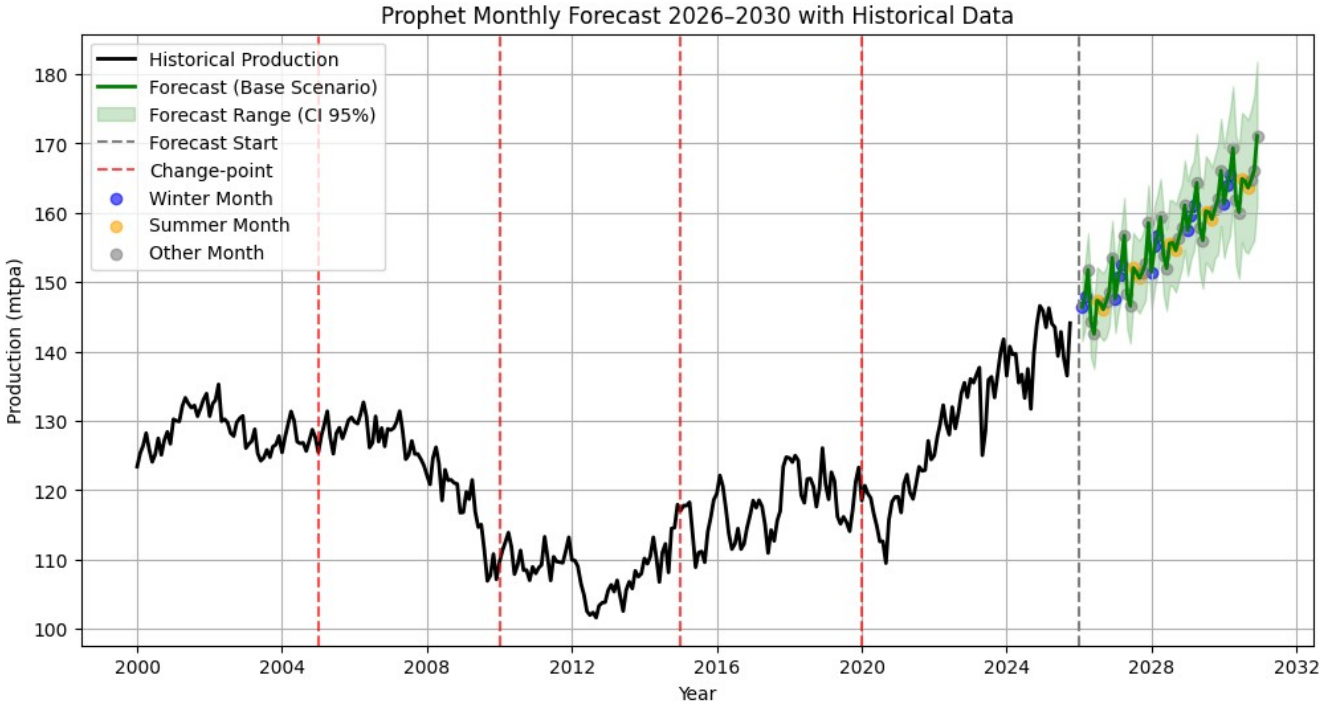

Figure 9: Prophet Monthly Forecast 2026–2030 with Historical Data

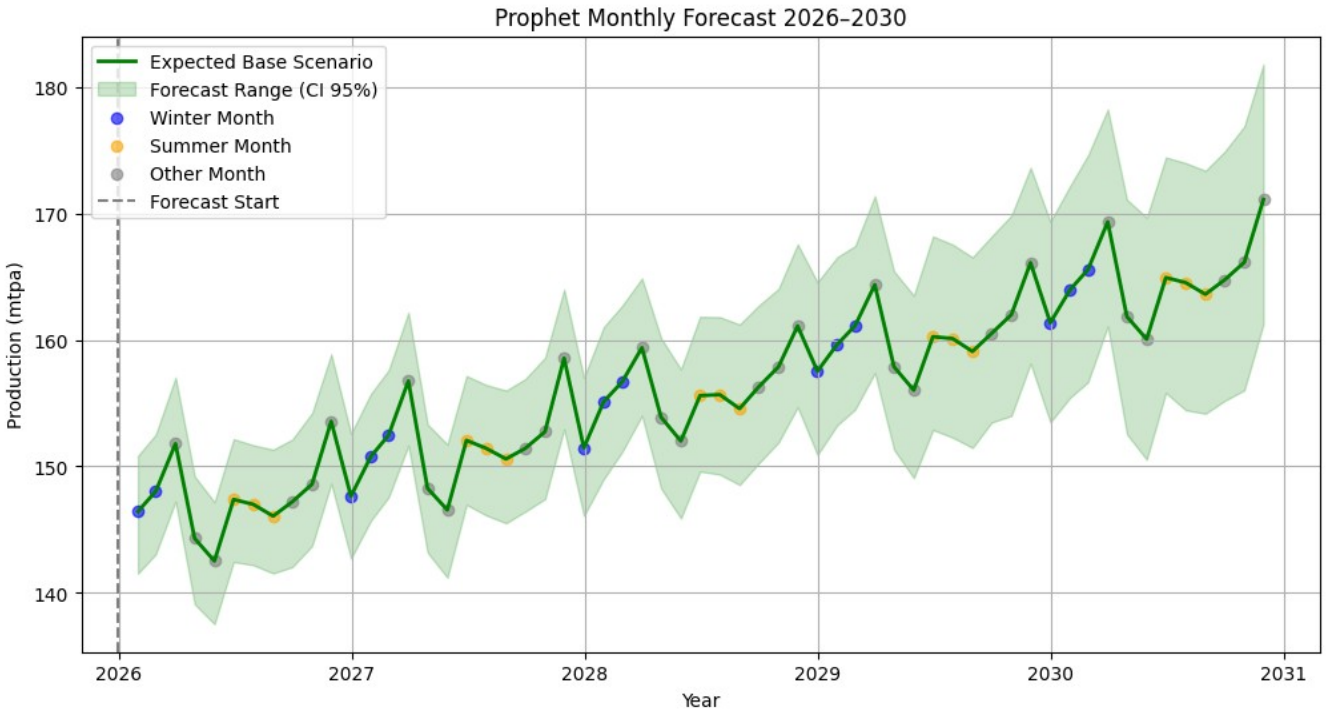

Figure 10: Prophet Monthly Forecast 2026–2030

Figure 9 presents the long-term forecast for 2026–2030 alongside historical production. The model projects a continued upward baseline trend following recent structural shifts. Seasonal patterns remain clearly embedded in the forecast, with winter months consistently forming the upper spread of the projection and summer months forming the lower trough.

Figure 10 focuses exclusively on the 2026–2030 forecast horizon. The seasonal distribution is clearly structured: winter spreads are systematically projected at peak levels, while summer months remain at relative lows, indicating that the model preserves historical seasonal amplitude in future projections. The smooth and orderly seasonal oscillation suggests that the model expects continued cyclical production behavior rather than structural disruption. Overall, the forecast indicates stable seasonal volatility layered on top of a gradually evolving trend.

7-2. SARIMA

The SARIMA model is an extension of ARIMA that incorporates seasonal patterns, making it suitable for capturing recurring monthly fluctuations in gas production. The model is specified with non- seasonal order (p,d,q) = (1,1,1) and seasonal order (P,D,Q,s) = (1,1,1,12), which allows it to flexibly capture both trend and seasonal changes. Stationarity and invertibility constraints were relaxed to improve model stability and accommodate potential structural shifts. Similar to Prophet, SARIMA leverages historical data to forecast future production, capturing seasonal peaks (Winter) and troughs (Summer) in a structured way.

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7-2-1. Validation

The SARIMA model's forecast closely follows the actual production values for 2023–2025, showing better alignment than the Prophet model. Seasonal patterns are clearly represented, with higher production in Winter and lower in Summer. The forecast also includes confidence intervals to indicate the expected range of variation.

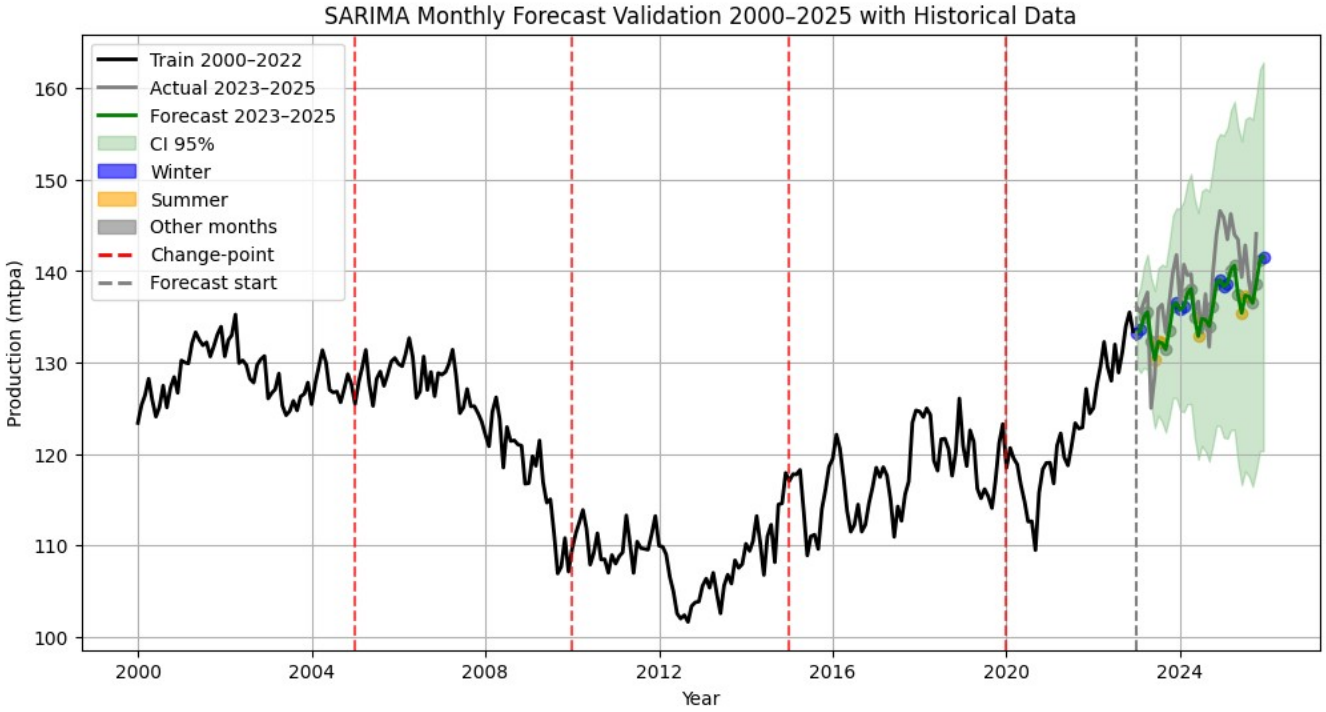

Figure 11: SARIMA Monthly Forecast Validation 2000–2025 with Historical Data

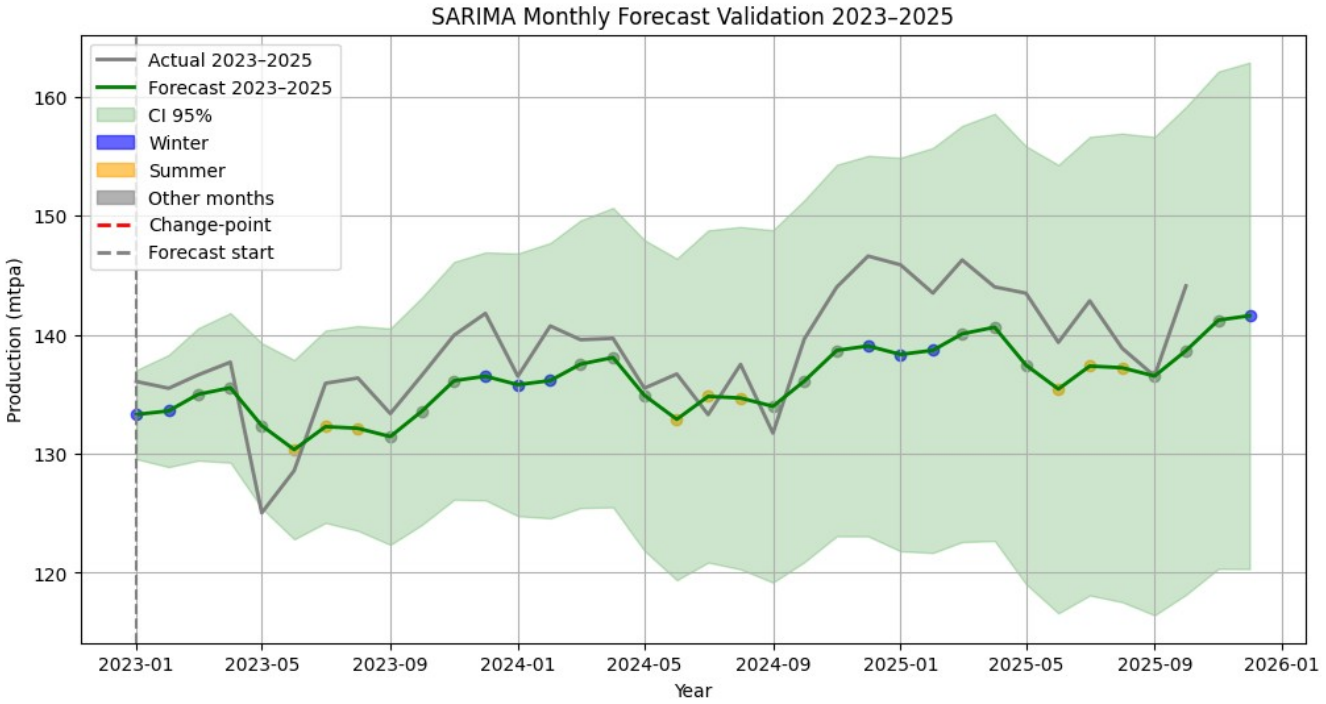

Figure 12: SARIMA Monthly Forecast Validation 2023–2025

7-2-2. Forecast

Figures 13 and 14 present the SARIMA forecast for 2026–2030. The model projects steady growth while preserving established seasonal cycles, with recurring Winter peaks and Summer troughs. Overall, production is expected to follow stable cyclical patterns alongside moderate upward momentum.

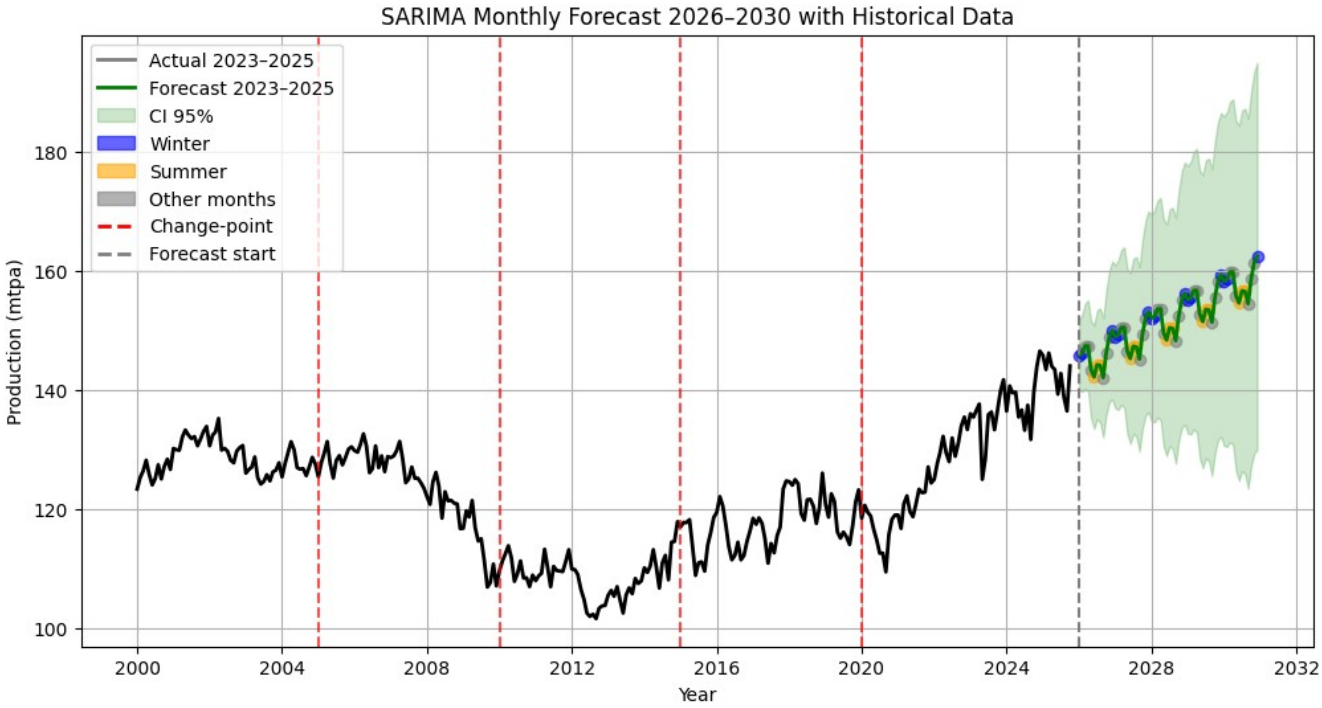

Figure 13: SARIMA Monthly Forecast 2026–2030 with Historical Data

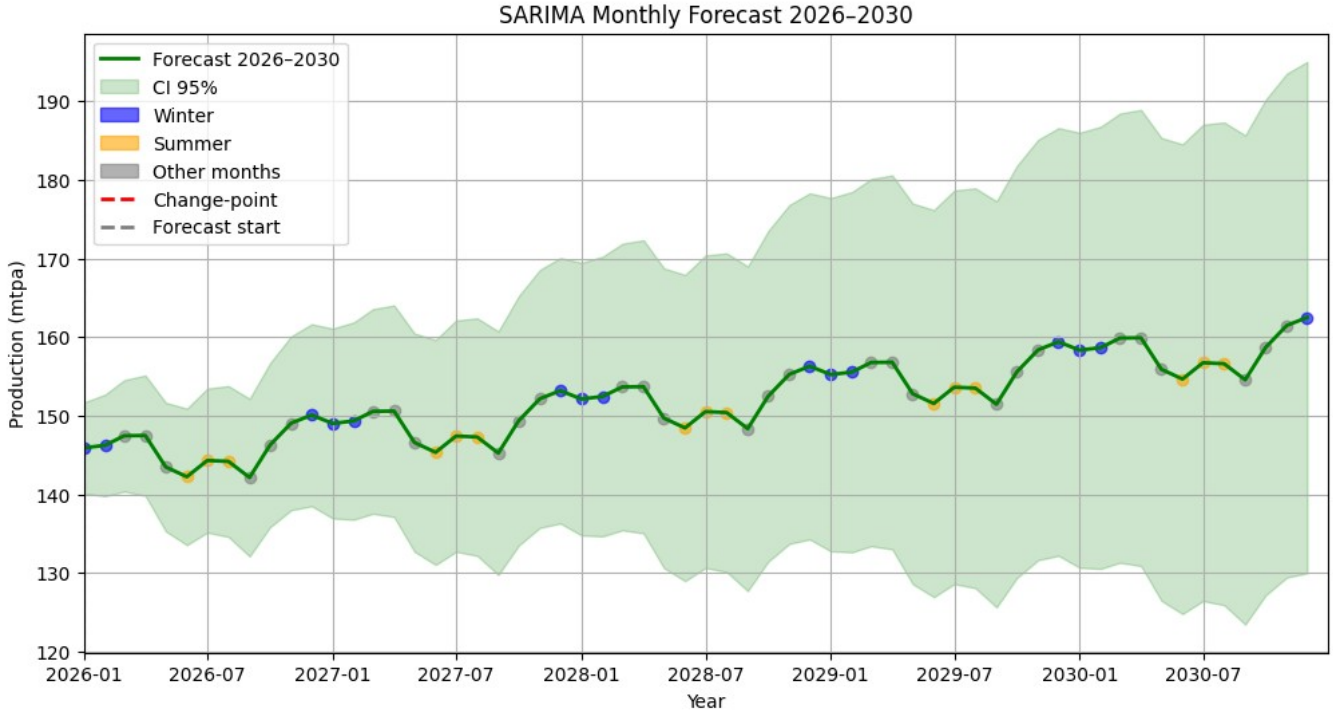

Figure 14: SARIMA Monthly Forecast 2026–2030

7-3. Incremental Supply vs LNG Canada Demand

7-3-1. Baseline Definition (2025 Production Level)

To conduct a partial equilibrium shock analysis, we define the 2025 total production level as the baseline. This baseline represents the pre-LNG equilibrium supply level prior to the demand shock from LNG Canada Phase 1. Future incremental supply is therefore measured as the difference between forecasted production (2026-2030) and the 2025 production level, allowing us to assess whether additional output is sufficient to meet 14 mtpa LNG requirement.

7-3-2. Incremental Supply Projection

Figure 15 presents the projected incremental gas surplus for 2026 to 2030 relative to the 2025 baseline. The results indicate that the incremental production increase remains below the 14 mtpa requirement of LNG Canada Phase 1 through 2029, with adequacy achieved only by 2030. This suggests a temporary period of structural supply tightness during the early years of LNG operations.

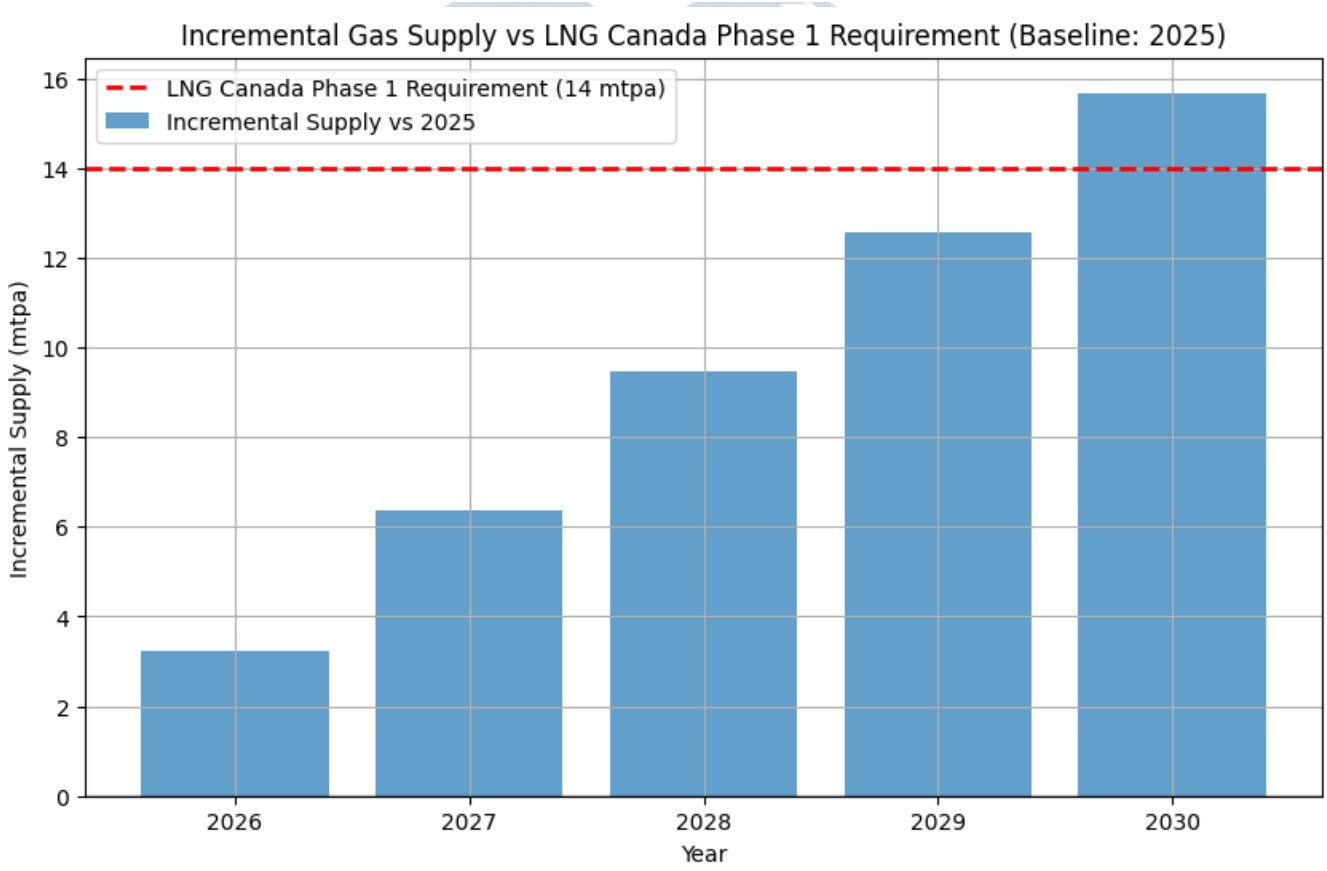

Figure 15: Incremental Gas Supply vs LNG Canada Phase 1 Requirement (Baseline 2025)

7-3-3. Supply Gap & Structural Tightness

Figure 16 presents the structural supply gap, defined as incremental supply minus the 14 mtpa LNG Canada Phase 1 requirement, for 2026–2030. By expressing the results in gap form, the analysis directly identifies period of deficit (negative values) and surplus (positive values). The transition from negative to positive territory marks the timing of structural balance, indicating a regime shift from supply tightness to adequacy.

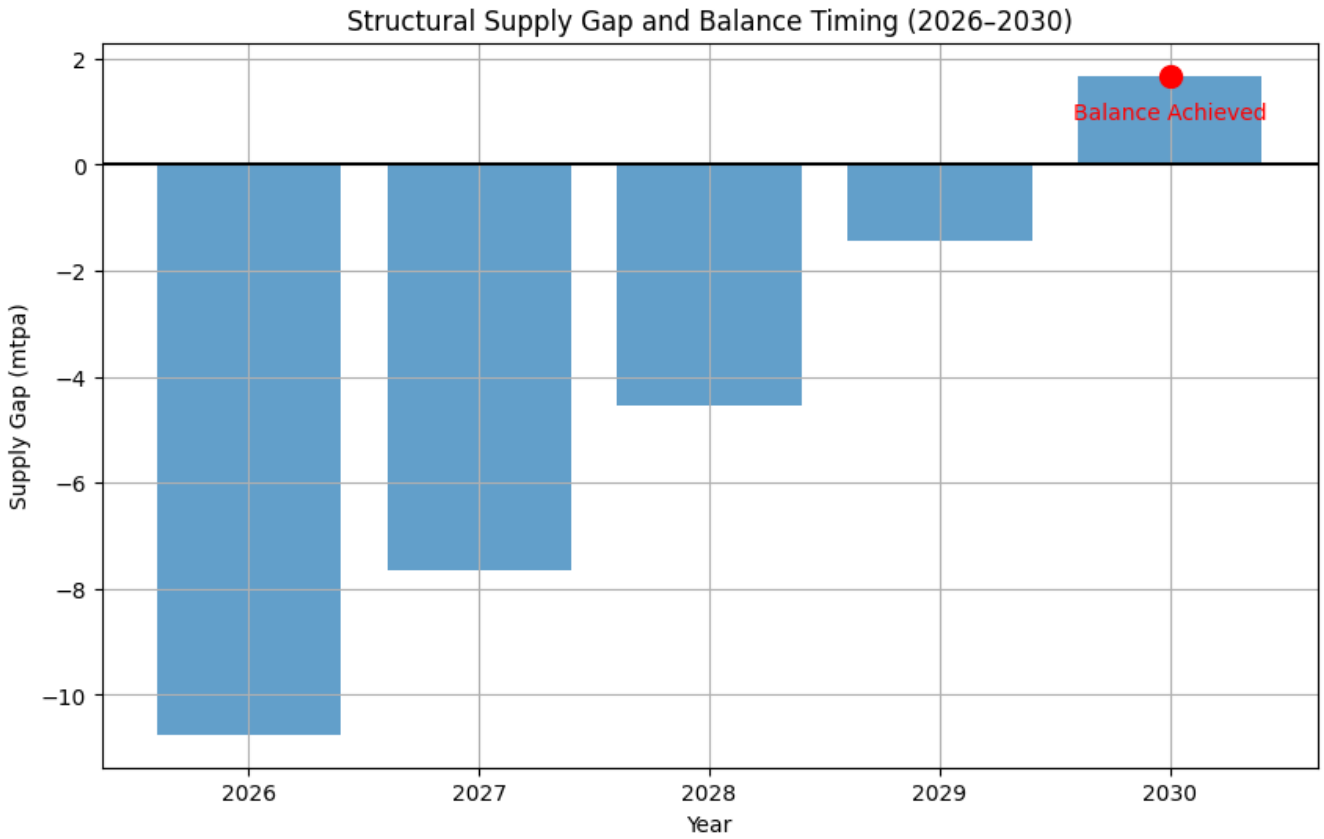

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Figure 16: Structural Supply Gap and Balance Timing (2026–2030)

8. Limitations & Uncertainty

While the Prophet and SARIMA models project production growth based on historical trends, they implicitly assume **unconstrained infrastructure capacity**. In reality, the effective delivery of incremental gas supply from the Western Canadian Sedimentary Basin (WCSB)[9][10] to the LNG export terminal depends on the available capacity of the Coastal GasLink pipeline. If takeaway capacity becomes binding, additional upstream production may not be fully transported to the coast, limiting realized LNG feedgas supply. Consequently, the projections in this study should be interpreted as unconstrained supply scenarios, and actual outcomes may be affected by midstream infrastructure constraints.

9. Conclusion

Overall, both historical trends and model projections indicate that Canada's upstream natural gas production has the potential to support LNG Canada Phase 1 over the medium term. Structural supply gap analysis suggests that incremental supply may remain insufficient during the initial years of LNG operations, with balance projected around 2030, indicating a transitional period of relative tightness.

While production growth appears stable and gradually increasing, the projections assume unconstrained pipeline and infrastructure capacity and do not account for midstream or downstream limitations, market dynamics, or operational bottlenecks. Therefore, the results should be interpreted as indicative of upstream production potential under baseline, unconstrained scenarios rather than as definitive outcomes of LNG exports.

10. References

10-1. Energy & Historical Events

[1] Environmental Health Project, *Understanding Shale Gas Development*, [Online]. Available: https://www.environmentalhealthproject.org/understanding-sgd. [Accessed: Feb. 11, 2026].

[2] The Breakthrough Institute, *History of the Shale Gas Revolution*, Dec. 14, 2021. [Online]. Available: https://thebreakthrough.org/issues/energy/history-of-the-shale-gas-revolution. [Accessed: Feb. 11, 2026].

[3] Export Development Canada (EDC), *COVID-19: NAVIGATING THE TURMOIL IN CANADA'S ENERGY SECTOR*, Apr. 17, 2020. [Online]. Available: https://www.edc.ca/content/dam/edc/en/non-premium/ungated/guide/covid-19-sector-oil-and-gas-crisis.pdf. [Accessed: Feb. 11, 2026].

[4] LNG Canada, *LNG Canada Announces a Positive Final Investment Decision*, Oct. 1, 2018. [Online]. Available: https://www.lngcanada.ca/news/lng-canada-announces-a-positive-final-investment-decision/. [Accessed: Feb. 11, 2026].

[5] World Bank Blogs, M. Stocker, J. Baffes, D. Vorisek, *What triggered the oil price plunge of 2014-2016 and why it failed to deliver an economic impetus in eight charts*, Jan. 18, 2018. [Online]. Available: https://blogs.worldbank.org/en/developmenttalk/what-triggered-oil-price-plunge-2014-2016-and-why-it-failed-deliver-economic-impetus-eight-charts. [Accessed: Feb. 11, 2026].

[6] LNG Canada, *About LNG Canada*. [Online]. Available:

https://www.lngcanada.ca/who-we-are/about-lng-canada/. [Accessed: Feb. 11, 2026].

[7] Shell, *First cargo leaves LNG Canada*, Jun. 30, 2025. [Online]. Available: https://www.shell.com/news-and-insights/newsroom/news-and-media-releases/2025/first-cargo-leaves-lng-canada.html. [Accessed: Feb. 11, 2026].

[8] Canada Energy Regulator, *Market Snapshot: Exploring Canada's Future in LNG Exports*, Sept. 4, 2024. [Online]. Available: https://www.cer-rec.gc.ca/en/data-analysis/energy-markets/market-snapshots/2024/market-snapshot-exploring-canadas-future-in-lng-exports.html. [Accessed: Feb. 11, 2026].

[9] British Columbia Energy Regulator, *Coastal GasLink*, Oct. 30, 2025. [Online]. Available: https://www.bc-er.ca/what-we-regulate/major-projects/coastal-gaslink/. [Accessed: Feb. 28, 2026]

[10] The British Columbia Environmental Assessment Office (EAO), *Coastal GasLink Pipeline*. [Online]. Available: https://www.projects.eao.gov.bc.ca/p/588511c4aaecd9001b825604/project-details. [Accessed: Feb. 28, 2026]

10-2. Forecasting Techniques & Python Resources

[11] Number Analytics, S. Lee, *Mastering Prophet for Time Series Forecasting*, May. 14, 2025. [Online]. Available: https://www.numberanalytics.com/blog/mastering-prophet-time-series. [Accessed: Feb. 25, 2026].

[12] Medium, I. Moore, *Time Series Forecasting Part 2: ARIMA vs SARIMA vs Holt-Winters on Stationary Data*, Sept. 8, 2025. [Online]. Available: https://medium.com/@ian.moore_83986/time-series-forecasting-part-2-arima-vs-sarima-vs-holt-winters-on-stationary-data-a6e222bbab25. [Accessed: Feb. 14, 2026].

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11. Glossary

[i] LNG – Liquefied Natural Gas

[ii] EDA – Exploratory Data Analysis

[iii] CI – Confidence Interval